0A Integration Roadmap

FIELD GUIDE

HOW TO USE THIS BINDER

Use one sheet at the bench. Check each box only after the action is complete. Record measurements in testing/test-matrix.csv. A HOLD badge means prerequisites are unresolved; do not improvise past it.

DEPENDENCY SPINE

1 **Complete boards and cables** →

2 **Prove PCB-01 without motor** →

3 **Measure bare GL100, no blades**

4 **Integrate and balance rotor** →

5 **Proof, starts, and thermal** →

CURRENT CHECKPOINT

- ☐ PCB-01 and PCB-02 hand-population complete and inspected.
- ☐ Power, motor, Hall, and programming harnesses built and continuity-checked.
- ☐ PCB-01 passes PCB-01 through PCB-04 and TACH-01 without the motor.

NON-NEGOTIABLE BOUNDARIES

- ☐ No blades during bare-motor characterization.
- ☐ No powered motor work before the no-motor board and safety-chain checks pass.
- ☐ Balance and hand-clearance checks pass before any powered full-rotor run.
- ☐ Use a guarded fixture and released procedure for the 216 RPM rotor proof.
- ☐ Never bypass the 180 RPM controller limit or 200 RPM analog trip without the written guarded two-person procedure.

1A Complete PCB-01 and PCB-02

FIELD GUIDE

TOOLS

ESD mat - microscope - temperature-controlled iron/hot air - flux - solder wick - DMM

PCB-01 HAND-POPULATION

- [ ] Inspect the assembled board for shipping damage and solder bridges.
- [ ] Fit C1 and C2: Panasonic FR 470 uF / 50 V. Observe polarity and silkscreen.
- [ ] Fit J1: Molex 43045-0200 power header.
- [ ] Fit J2: Molex 43650-0300 phase header.
- [ ] Fit U8: LM2907M/NOPB. Match pin 1 to board marking.
- [ ] Bridge C34 with KEMET C1206C104K3GACTU, 100 nF C0G, across the 0603 site.
- [ ] Bridge F1 pads. The wall-side 3 A fuse is then mandatory.
- [ ] Inspect every hand joint and verify no RAW24-to-ground short.

PCB-02 HAND-ASSEMBLY

- [ ] Fit U1: DRV5033FAQDBZR, with package orientation matching PCB marking.
- [ ] Fit C1: 100 nF / 50 V X7R 0603 local bypass.
- [ ] Fit J1: S3B-PH-K-S side-entry JST-PH.
- [ ] Keep every component on the magnet-facing side; solder before mounting to BR-100.
- [ ] Inspect under magnification and continuity-check 3V3, HALL_TACH, and AGND.

DO NOT POWER either board until 1B harness polarity and the unpowered continuity checks are complete.

STOP / HANDOFF Both boards are fully populated, photographed, and pass unpowered short/polarity inspection.

SOURCE: docs/electrical.md, PCB-01 mechanical definition and SCH-06; bom/bom.csv, PCB fab rows

1B Build and Verify Harnesses

FIELD GUIDE

CONNECTOR TRUTH

CONNECTOR	FIRST PINS	REMAINDER
J1 POWER	1 RAW24	2 0V
J2 MOTOR	1 W	2 V | 3 U
J3 / PCB-02 J1	1 3V3	2 HALL_TACH | 3 AGND
J5 I2C	GND, 3V3	SDA, SCL
J7 PROGRAM	3V3, TX, RX	EN, BOOT, GND

BUILD CHECKLIST

- [ ] Power harness: Belden 5300UE 18/2 to Micro-Fit; preserve +24 V and 0 V end-to-end.
- [ ] Motor harness: use 18 AWG contacts; label connector positions W, V, U even if motor flying leads are initially unlabeled.
- [ ] Hall harness: PHR-3 to PHR-3 straight-through, using molded No.1 marks at both ends.
- [ ] Programming lead: J7 pin 1 is board 3V3, not a power input.
- [ ] Pull-test every crimp, then continuity-check each contact and verify no cross-continuity.
- [ ] Record harness length, wire colors, and contact orientation before sleeving.

J7 pin 1 ties directly to board 3V3. Never connect a programmer that actively drives 3.3 V while 24 V is connected.

A swapped Hall contact can leave the tach silently dead. Require 1-to-1, 2-to-2, 3-to-3 and no cross-continuity before first power.

STOP / HANDOFF Every harness is labeled, pull-tested, and has a signed continuity result. No device is powered yet.

1C PCB-01 First Power and Safety Chain

FIELD GUIDE

SAFE TEST SETUP

- ☐ Disconnect motor phases and keep the rotor absent.
- ☐ Use the fused 24 V bench path with current limiting and a physical cutoff within reach.
- ☐ Connect console/programmer without driving board 3V3 from J7.
- ☐ Place probes before power; use spring ground for VM transient measurements.

CHECK IN THIS ORDER

- ☐ PCB-01: power briefly; verify 24 V, 3.3 V, AVDD, DVDD, DRVOFF, current draw, and no abnormal heating.
- ☐ PCB-03: force PGOOD, WDO, manual clear, and OVERSPEED_N low one at a time. DRVOFF must rise without ESP help.
- ☐ PCB-03B: static WDI must time out in 1.44-1.76 s; WDO pulse about 200 ms. Falling edges at 2 Hz prevent timeout.
- ☐ PCB-03D: repeat cold and slow-ramp power-up 20 times; U6 /PRE must release after TACH_PGOOD_N.
- ☐ PCB-04: remove and restore 24 V. Fan remains disabled for at least 10 s and until a new command plus explicit arm.
- ☐ TACH-01: inject 0-3.3 V at HALL_TACH with bridge disabled, 1-3% duty, and at least 30 s settling.
- ☐ Record reset near 3.00 Hz and trip near 3.33 Hz; the persistent lock resets only on genuine power cycle.

PASS LIMITS

VM insertion target: <=35 V; coast/cutoff/stall/reversal checks occur later with the guarded motor. Watchdog: 1.44-1.76 s. Power-up: 20/20 correct presets. Tach: reset about 3.00 Hz, trip about 3.33 Hz. Any automatic re-arm blocks release.

STOP / HANDOFF PCB-01 through PCB-04 and TACH-01 have recorded pass/fail results. Do not connect the motor after a failure.

3A Stationary Stack Dry-Fit

FIELD GUIDE

STACK ORDER

MP-100 CEILING PLATE

↓

3 x ST-100, 62 mm

↓

MC-100 CARRIER

↓

GL100 STATIONARY FACE

Fasteners

- [ ] MP-100 to ST-100: 3 x M6 x 16 flat-head from ceiling face.
- [ ] MC-100 to ST-100: 3 x M6 x 20 A4-80 with wedge-lock pairs.
- [ ] MC-100 to GL100 rear face: 4 x M4 x 12 A4-80 on 60 mm PCD.

DRY-FIT CHECKLIST

- [ ] Confirm GL100 stationary rear face is the 60 mm M4 pattern; rotating output is the 50 mm pattern.
- [ ] Confirm the motor wire exit clears the MC-100 phase window without pinching.
- [ ] Hand-start every fastener. Stop if any screw bottoms before clamping.
- [ ] Verify the three ST-100s seat squarely and MC-100 is not rocked by finish or burrs.
- [ ] Verify M4 x 12 gives about 5.5 mm engagement and never exceeds the GL100 6.0 mm limit.
- [ ] Use the owner-selected assembly torque consistently with a calibrated driver.
- [ ] Apply compatible removable threadlocker and witness marks at final assembly.
- [ ] Record stack fit, wire clearance, and any washer/length adaptation before proceeding.

Torque research is closed. Use the owner-selected value consistently, then witness-mark each fastener.

STOP / HANDOFF The stationary stack fits without force, rocking, wire pinch, or bottoming screws. Keep it non-powered.

SOURCE: docs/parts.md, GL100/ST-100/MC-100 and Fastener release practice; bom/bom.csv

3B Rotor and Tach Inserts

FIELD GUIDE

ROTOR ASSEMBLY

- [ ] RH-100 pilot enters the GL100 bore by hand; never press or force it.
- [ ] Install RH-100 with 4 x M4 x 10 A4 flat-head screws from the underside into the 50 mm rotating pattern.
- [ ] Verify every hub screw head is 0.1-0.2 mm subflush.
- [ ] Fit one magnet and two mass-matched brass slugs in the three r76 pockets; epoxy is the retention.
- [ ] Match each complete insert-plus-epoxy station within 0.01 g.
- [ ] Install three BP-100 v3 blades using their printed locating pins and 4 x M5 x 22 bolts per blade into the captured M5 nuts.

ROTOR FINAL CHECKS

- [ ] Confirm all three blade stations use the correct fasteners and captured nuts.
- [ ] Verify the magnet and both brass stations are fully retained after epoxy cure.
- [ ] Apply the owner-selected torque and witness marks to hub and blade fasteners.
- [ ] Hand-rotate several revolutions and confirm no rub, click, or changing clearance.
- [ ] Photograph the completed rotor before balance measurements begin.

BEFORE ANY POWER

- [ ] Hand-rotate through multiple revolutions; confirm no rub, click, or changing clearance.
- [ ] Verify all critical fasteners have consistent torque and witness marks.
- [ ] Do not use cad/BP-100.step for current fit; the committed export is stale v2 geometry.
- [ ] Proceed to balance and runout measurement before powered full-rotor testing.

STOP / HANDOFF The complete rotor turns freely by hand with correct retention and no contact.

SOURCE: docs/parts.md, RH-100 and tach inserts; docs/blade-v2.md

3C Hall and Electronics Bracket

FIELD GUIDE

HALL SENSOR GEOMETRY - RELEASED

- [ ] PCB-02 components face the rotor magnet; J1 and cable exit outboard.
- [ ] Sensor element and magnet centerline are both at radius 76.0 +/- 0.5 mm.
- [ ] Set 2.5 mm nominal axial gap, adjustable 1.5-4.0 mm, measured from the marked SOT-23 outer face to magnet cap.
- [ ] Use PCB-02 M2 holes on 6 mm pitch. At H1 use a bare pan head or washer <=4.5 mm OD.
- [ ] Provide cable strain relief on BR-100; PCB material does not extend beyond J1.
- [ ] Hand-turn rotor: one clean pulse per revolution with magnet; no false pulses with magnet removed.

EB-100 AFTER CONNECTORS

- [ ] BR-100 remains owner hand-fabricated; validate around the physical motor, PCB-02, cable, and Hall gap.
- [ ] Populate PCB-01 connectors before defining EB-100 and real cable bends.
- [ ] Reserve 110 x 80 x 25 mm for PCB-01 plus connector and service clearance.
- [ ] Use four 6-8 mm M3 standoffs and keep isolated mounting holes clear of circuit ground.
- [ ] Provide independent PCB retention and clamp cables independently of the bracket.

RELEASE CHECK

- [ ] Connector bodies and cable bends clear the rotor and standoffs.
- [ ] PCB-01 has an independent retention lanyard and independent cable clamps.
- [ ] Hand rotation remains free after the bracket and all cables are fitted.

STOP / HANDOFF Hall pulses are clean and the retained PCB/cables clear every moving part.

4A Flash, Console, and CLI Session

FIELD GUIDE

BUILD AND FLASH

```
cd firmware && cargo build
```

```
espflash flash --port /dev/cu.usbmodem2101 --non-interactive
  app/target/riscv32imac-unknown-none-elf/debug/stillair
```

```
target/debug/stillair --port /dev/cu.usbmodem2101 state
```

BARE DEV-BOARD INPUT SIMULATION

- [ ] GPIO22 to 3V3: PGOOD good.
- [ ] GPIO21 to 3V3: nFAULT idle high.
- [ ] GPIO14 to GND: ALARM idle low.
- [ ] Without real FG edges, Starting must fault NoRotation after 15 s. This is expected.

SESSION RULES

- [ ] Use one serial reader at a time. Stop raw-log capture before espflash or the CLI opens the port.
- [ ] Use `script` for any multi-step sequence. Separate CLI invocations create separate simulator sessions.
- [ ] Use `wait <state> --for <seconds>` after asynchronous commands.
- [ ] Use `stream <hz> --for <seconds>` for CSV telemetry.
- [ ] For release, `config check` must report config=verified; ok=true with config=unverified is not a pass.
- [ ] Use `config capture` only after a complete measured configuration exists.
- [ ] Treat simulator results as harness validation, never motor validation.

WARNING: The CLI only surfaces @-prefixed protocol lines. Capture raw logs for Wi-Fi/Matter failures. Never use raw reg write on a spinning motor except under an approved stopped/instrumented procedure.

STOP / HANDOFF Firmware flashes, one CLI session communicates reliably, and scripted sequences preserve device state.

SOURCE: AGENTS.md, Driving the fan; firmware/cli/src/main.rs

4B Bare-Motor Measurements and Config Gate

HOLD: IMAGE UNVERIFIED

PREREQUISITES

- [ ] PCB-01 through PCB-04 and TACH-01 passed.
- [ ] GL100 is rigidly guarded with NO BLADES installed.
- [ ] VM scope probe and manual cutoff are ready before power.

BARE-MOTOR WORK ALLOWED NOW

- [ ] Measure phase R and L independently with the method and connection convention recorded.
- [ ] Manually spin the GL100 and scope line-to-line BEMF; record amplitude and phase convention.
- [ ] Record the measured values beside provisional seeds 0xB1 / 0xAE / 0xCA. Do not call the seeds released.
- [ ] Confirm 20 pole pairs and inspect console, FG, Hall, watchdog, and stop behavior without raw live register writes.
- [ ] Scope only the motor operations already approved by the guarded procedure; VM target <=35 V and 40 V rejects.
- [ ] Keep blades and loose hub hardware off throughout this bare-motor stage.

MPET AND GOLDEN IMAGE - HOLD

- [ ] Do not run MPET unloaded. controls.md requires the representative final rotor because unloaded MPET can produce bad Ke/Kp/Ki.
- [ ] Resolve the representative-rotor MPET procedure, then cross-check it against independent R/L/BEMF measurements.
- [ ] Populate the required D-generation fields, including safety-critical speed, ALARM, and external-watchdog settings.
- [ ] Do not use `config apply` as a release step until EEPROM completion timing and readback are proven on hardware.
- [ ] Capture the complete image only after review. Never reuse A1/C-generation register dumps.
- [ ] Require `config check` to report config=verified; ok=true with config=unverified is not a release pass.

CURRENT RECORD

R: __________ **L:** __________ **BEMF:** __________ **convention:** __________________________

Config verdict: unverified / verified image commit: ______________________________

STOP / HANDOFF Independent bare-motor measurements are recorded. MPET and the golden image remain HOLD until the loaded procedure is released.

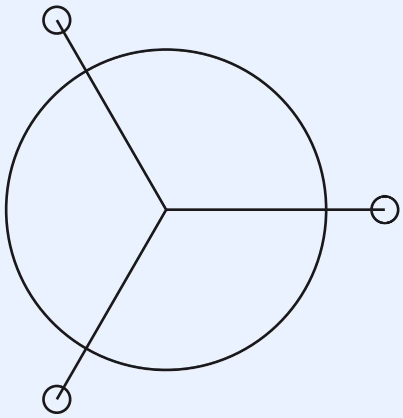
5A Rotor Balance and Runout

FIELD GUIDE

MEASURE BEFORE POWER

- ☐ Use a released level/indexed fixture, a dial indicator readable to 0.01 mm, and a calibrated scale readable to 0.01 g.
- ☐ Define the vibration measurement and rejection criterion before any powered motion.
- ☐ Record each blade mass and first moment; target first moments within 0.5%.
- ☐ Measure RH-100 OD runout; limit <=0.10 mm TIR.
- ☐ Measure all blade tips in one indexed setup; spread <=0.5 mm.
- ☐ Hand-rotate and verify no rotating-to-stationary contact.
- ☐ Correct only with the documented balance-slug method; re-measure after every change.
- ☐ Photograph witness marks and record final correction masses/locations.

VISUAL RECORD



Hub TIR: ______ **mm**

Tip spread: ______ **mm**

First-moment spread: ______ **%**

Correction: ______________________

PASS CONDITION

Hub <=0.10 mm TIR, tips within 0.5 mm, first moments within 0.5%, no objectionable vibration, and no contact.

STOP / HANDOFF MEC-05 passes with measurements recorded. Do not advance a rotor that only 'looks balanced.'

5B Guarded Rotor Proof

HOLD: PROCEDURE MISSING

ENGINEER BEFORE TESTING

- [ ] Release a fixture drawing with rated containment, external drive, remote cutoff/interlock, and instrument mounting.
- [ ] Write the two-person procedure: roles, callouts, abort criteria, maximum duration, safe positions, and restoration checks.
- [ ] Define the exact bypass hardware/state if a temporary electronic bypass is unavoidable.
- [ ] Provide independent RPM measurement and a measurable vibration/contact abort criterion.
- [ ] Record pre-run balance, runout, fastener witness marks, and hand-clearance condition.
- [ ] Approve the procedure before any high-energy run.

PROOF REQUIREMENTS

- [ ] MEC-03 proof requirement: external guarded drive, 216 RPM, 2 minutes each direction.
- [ ] After any bypass, restore and independently reverify the 180 RPM MCF limit and 200 RPM analog trip before fixture removal.
- [ ] Never dynamically test the 270 RPM credible bypass load. It is calculation-only.
- [ ] Use the manual cutoff immediately for unexpected vibration, sound, contact, or speed deviation.
- [ ] Reject damage, opening, permanent set, deformation, balance shift, contact, or loosened witness marks.
- [ ] Repeat balance, runout, witness-mark, and hand-clearance checks after both directions pass.

STOP / HANDOFF No test is authorized by this sheet. Release the fixture and procedures first; then issue an execution revision.

5C Representative Starts and Thermal

HOLD: METHODS TO DEFINE

START AND SPEED QUALIFICATION

- [ ] With the representative final rotor guarded, run the released MPET procedure; cross-check R/L/BEMF, then complete and read back the golden image.
- [ ] Before starts, require config check to report config=verified and independently verify the 180 RPM MCF limit and 200 RPM analog trip.
- [ ] At the intended minimum, complete 20 randomized-rest starts per direction at 24.0 V.
- [ ] At the same minimum, complete 5 starts per direction at 23.3 V and 5 per direction at 24.7 V.
- [ ] Reject any retry, reverse kick, stall, hunting, or objectionable tonal sequence.
- [ ] Verify steady operation at 30, 40, 55, 70, 120, and 170 RPM.
- [ ] Verify MCF active-control ceiling never exceeds 180 RPM.
- [ ] Test reversal after verified stop, watchdog shutdown, power recovery, and cutoff at speed.

THERMAL

- [ ] Record ambient condition, temperature-sensor locations, and logging cadence.
- [ ] Run the complete fan for 8 hours at 170 RPM on the guarded fixture.
- [ ] Record RMS phase current, MCF q-axis estimate, PWM peak, independent motor/PCB temperatures, ambient, and supply behavior.
- [ ] Normal RMS phase current <0.8 A; investigate at 1.0 A; 1.5 A limiter must not clip continuously.
- [ ] Motor <70 C, PCB <85 C, no supply dropout or overtemperature, input power <50 W.
- [ ] After cooldown, repeat hand-clearance and fastener witness-mark checks.

RELEASE SUMMARY

Released minimum RPM: ______ Max qualified RPM: ______

Open anomalies / restrictions: ____

STOP / HANDOFF Representative starts, stable speeds, essential shutdowns, and the thermal run pass.